

Horizontal Polarization

$$\mathbf{E}_h^i = \hat{\mathbf{y}} E_{h0}^i e^{-jk_1(x \sin \theta_1 - z \cos \theta_1)}$$

$$\mathbf{H}_h^i = (\hat{\mathbf{x}} \cos \theta_1 + \hat{\mathbf{z}} \sin \theta_1) \frac{E_{h0}^i}{\eta_1} e^{-jk_1(x \sin \theta_1 - z \cos \theta_1)}$$

$$\mathbf{E}_h^r = \hat{\mathbf{y}} \rho_h E_{h0}^i e^{-jk_1(x \sin \theta_1 + z \cos \theta_1)}$$

$$\mathbf{H}_h^r = (-\hat{\mathbf{x}} \cos \theta_1 + \hat{\mathbf{z}} \sin \theta_1) \rho_h \frac{E_{h0}^i}{\eta_1} e^{-jk_1(x \sin \theta_1 + z \cos \theta_1)}$$

$$\mathbf{E}_h^t = \hat{\mathbf{y}} \tau_h E_{h0}^i e^{-jk_2(x \sin \theta_2 - z \cos \theta_2)}$$

$$\mathbf{H}_h^t = (\hat{\mathbf{x}} \cos \theta_2 + \hat{\mathbf{z}} \sin \theta_2) \tau_h \frac{E_{h0}^i}{\eta_2} e^{-jk_2(x \sin \theta_2 - z \cos \theta_2)}$$

$$\rho_h = \frac{\eta_2 \cos \theta_1 - \eta_1 \cos \theta_2}{\eta_2 \cos \theta_1 + \eta_1 \cos \theta_2} = \frac{\cos \theta_1 - \sqrt{(\varepsilon'_2/\varepsilon'_1) - \sin^2 \theta_1}}{\cos \theta_1 + \sqrt{(\varepsilon'_2/\varepsilon'_1) - \sin^2 \theta_1}}$$

$$\tau_h = 1 + \rho_h$$

Vertical Polarization

$$\mathbf{E}_v^i = (-\hat{\mathbf{x}} \cos \theta_1 - \hat{\mathbf{z}} \sin \theta_1) E_{v0}^i e^{-jk_1(x \sin \theta_1 - z \cos \theta_1)}$$

$$\mathbf{H}_v^i = \hat{\mathbf{y}} \frac{E_{v0}^i}{\eta_1} e^{-jk_1(x \sin \theta_1 - z \cos \theta_1)}$$

$$\mathbf{E}_v^r = (\hat{\mathbf{x}} \cos \theta_1 - \hat{\mathbf{z}} \sin \theta_1) \rho_v E_{v0}^i e^{-jk_1(x \sin \theta_1 + z \cos \theta_1)}$$

$$\mathbf{H}_v^r = \hat{\mathbf{y}} \rho_v \frac{E_{v0}^i}{\eta_1} e^{-jk_1(x \sin \theta_1 + z \cos \theta_1)}$$

$$\mathbf{E}_v^t = (-\hat{\mathbf{x}} \cos \theta_2 - \hat{\mathbf{z}} \sin \theta_2) \tau_v E_{v0}^i e^{-jk_2(x \sin \theta_2 - z \cos \theta_2)}$$

$$\mathbf{H}_v^t = \hat{\mathbf{y}} \tau_v \frac{E_{v0}^i}{\eta_1} e^{-jk_2(x \sin \theta_2 - z \cos \theta_2)}$$

$$\rho_v = \frac{\eta_2 \cos \theta_2 - \eta_1 \cos \theta_1}{\eta_2 \cos \theta_2 + \eta_1 \cos \theta_1} = \frac{\left[\frac{\varepsilon'_2}{\varepsilon'_1} - \sin^2 \theta_1 \right]^{1/2} - \left(\frac{\varepsilon'_2}{\varepsilon'_1} \right) \cos \theta_1}{\left[\frac{\varepsilon'_2}{\varepsilon'_1} - \sin^2 \theta_1 \right]^{1/2} + \left(\frac{\varepsilon'_2}{\varepsilon'_1} \right) \cos \theta_1}$$

$$\tau_v = (1 + \rho_v) \frac{\cos \theta_1}{\cos \theta_2}$$

$$k_1 = \frac{2\pi}{\lambda_0} \sqrt{\varepsilon_1}$$

$$\eta_1 = \frac{\eta_0}{\sqrt{\varepsilon_1}}$$

$$\eta_2 = \frac{\eta_0}{\sqrt{\varepsilon_2}}$$

$$k_2 = \frac{2\pi}{\lambda} \sqrt{\varepsilon_2}$$

$$k_2 \sin \theta_2 = k_1 \sin \theta_1$$

$$\cos \theta_2 = \left[1 - \left(\frac{k_1}{k_2} \sin \theta_1 \right)^2 \right]^{1/2}$$

*Notes: (1) Lossless medium: $\varepsilon = \varepsilon'$; (2) lossy medium: $\varepsilon = \varepsilon' - j\varepsilon''$ and jk should be replaced with $\gamma = j(2\pi\sqrt{\varepsilon' - j\varepsilon''})/\lambda_0$.